Proof-Based Math Readings
Session: Linear Algebra*

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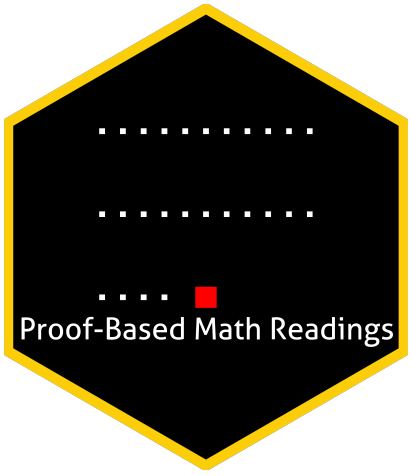
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0 Motivation

- *Proof-Based Math Readings* is a free, independent online reading group where we study the mathematics required for economics master's and PhD programs through an intuitive approach. Active since May 2023.
- This session of the reading group is on *Linear Algebra*.
- This session is dedicated to Sheldon Axler's lovely cat, Moon, who passed away in August 2023.

1 Prerequisites and Format

- Proof Techniques resources below and Linear Algebra - Gilbert Strang (2005).
- Please use the Application Form to join our reading group; you will receive a response within a week.
- This session takes 12 weeks. We do not have face-to-face/online meetings due to the size of the group.
- Members read the main book and discuss the topics/exercises in the Proof-Based Math Readings Discord .

2 Resources [All are open-access]

2.1 Main Book and Main Book's Playlist

Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Errata-free version) is our main book for this session because it is well-written, well-structured, and open-access.

Robert Won's playlist is our main playlist because his narrative is great.

- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Errata-free version)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Playlist by Robert Won)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Notes by Robert Won)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Playlist by Sheldon Axler)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Solutions by MathwithoutCommentary)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Solutions by Oliver Li)
- Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026, Solutions by nehc0)
- Linear Algebra Done Right - Sheldon Axler (3rd Edition, 2015, Solutions by jubnoske08)
- Linear Algebra Done Right - Sheldon Axler (3rd Edition, 2015, Solutions by Solverer)
- Linear Algebra Done Right - Sheldon Axler (3rd Edition, 2015, Playlist by Jason Morton)

2.2 Supplementary

2.2.1 Linear Algebra

- Essence of Linear Algebra - 3Blue1Brown (2023)
- Down with Determinants! - Sheldon Axler (1994)

2.2.2 Proof Techniques

- Book of Proof - Richard Hammack (3.4 Edition, 2025)
- Book of Proof - Richard Hammack (3.4 Edition, 2025, Playlist by Jeremy Teitelbaum)
- Book of Proof - Richard Hammack (3.4 Edition, 2025, Playlist by Michael Penn)

3 Reading Schedule

- **LADR** is the abbreviation of **Linear Algebra Done Right - Sheldon Axler (4th Edition, 2026)**.

LADR, Chapter 1: Vector Spaces — **Week 01**

1A $\mathbf{R}^n$ and $\mathbf{C}^n$
1B Definition of Vector Space
1C Subspaces

LADR, Chapter 2: Finite-Dimensional Vector Spaces — **Week 02-03**

2A Span and Linear Independence
2B Bases
2C Dimension

LADR, Chapter 3: Linear Maps — **Week 04-05-06**

3A Vector Space of Linear Maps
3B Null Spaces and Ranges
3C Matrices
3D Invertibility and Isomorphisms

LADR, Chapter 5: Eigenvalues and Eigenvectors — **Week 07-08**

5A Invariant Subspaces
5B The Minimal Polynomial
5C Upper-Triangular Matrices
5D Diagonalizable Operators

LADR, Chapter 6: Inner Product Spaces — **Week 09-10**

6A Inner Products and Norms
6B Orthonormal Bases
6C Orthogonal Complements and Minimization Problems

LADR, Chapter 7: Operators on Inner Product Spaces — **Week 11-12**

7A Self-Adjoint and Normal Operators
7B Spectral Theorem
7C Positive Operators
7D Isometries, Unitary Operators, and Matrix Factorization
7E Singular Value Decomposition

4 Further Readings (Optional)

Matrix Analysis - Roger A. Horn, Charles R. Johnson (2nd Edition, 2013)